

PROJECT DESIGN PHASE

Date	12 June 2026
Team ID	
Project Name	E-commerce Application (ShopEZ)
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. For the ShopEZ E-commerce Application, it establishes the operational blueprint aligning business workflows like inventory management, checkout, and client interactions with a scalable multi-tier software infrastructure.

Its goals are to:

- Find the best tech solution to solve existing business problems, specifically providing low-latency product indexing, consistency in inventory counts under peak loads, and robust transaction handling.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders, clarifying the decoupling of front-end storefront services from core relational inventory databases.
- Define features, development phases, and solution requirements, ensuring the iterative delivery of catalog browsing, cart operations, payment gateway integration, and back-office tracking modules.
- Provide specifications according to which the solution is defined, managed, and delivered, offering explicit guidelines for database connection pooling, REST API schemas, and security protocols during transaction handling.

Example - Solution Architecture Diagram :

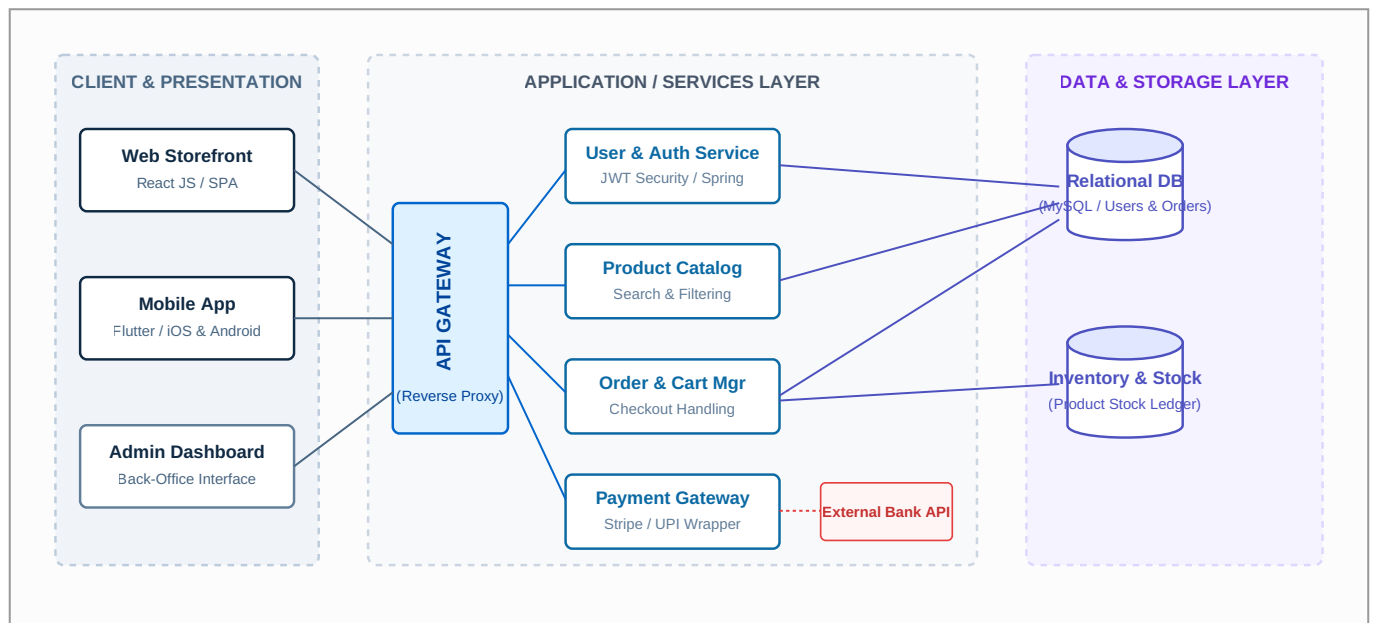


Figure 1: Architecture and data flow of the ShopEZ e-commerce sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>